# End to End Business Case →

## Agenda →

- Unsupervised ⇒ 3 Types
  - Dim^n Reduc^n
    - PCA
    - TSNE
    - Umap
    - viz^n
  - Clustering
    - kmeans
    - GMMs
    - Hierarchical
    - DBSCAN
  - Anomaly detection
    - iso
    - LOF

Business case → Supervised

- How price predⁿ

Preprocessing → Baseline score → improve

- RMSE, MAPE
- Unsupervised

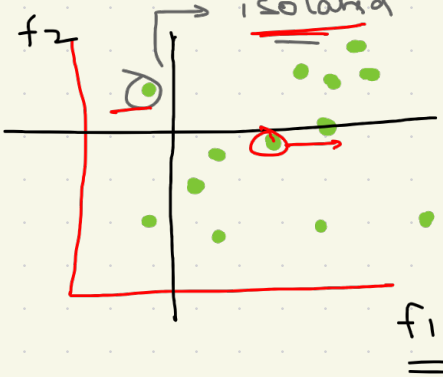
2) isolation forest

↓

Random Trees

$f_1$ or $f_2$ —

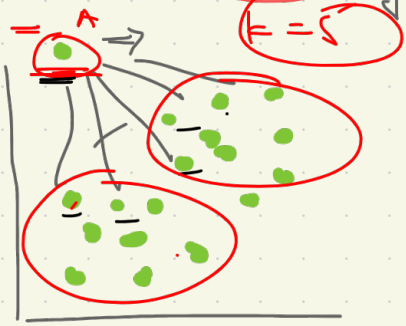
$\tau_1$ or $\tau_2$ —




4) LOF

KNN + density

$k = 5$




- $k^{th}$ distance

$5^{th}$ distance

(*) $N_k(A) \simeq N(A)$

set of Neighbours of A

$\{x^2, x^5, x^9\}$

3) RD

Reachability dist

$(A, B)$

$= \max(\text{dist}(A,B), \text{kn dist}(B))$

(4) LRD → [ ]$^{-1}$

High^n LRD (A) → dense spot

(5) LOF Score → $\frac{\text{avg}(\text{lrd}(\text{neighbors}))}{\text{lrd}(A)}$

$\approx 1$ ⇒ (A) dense inlier

$> 1$ → outlier

Payments company — card txns

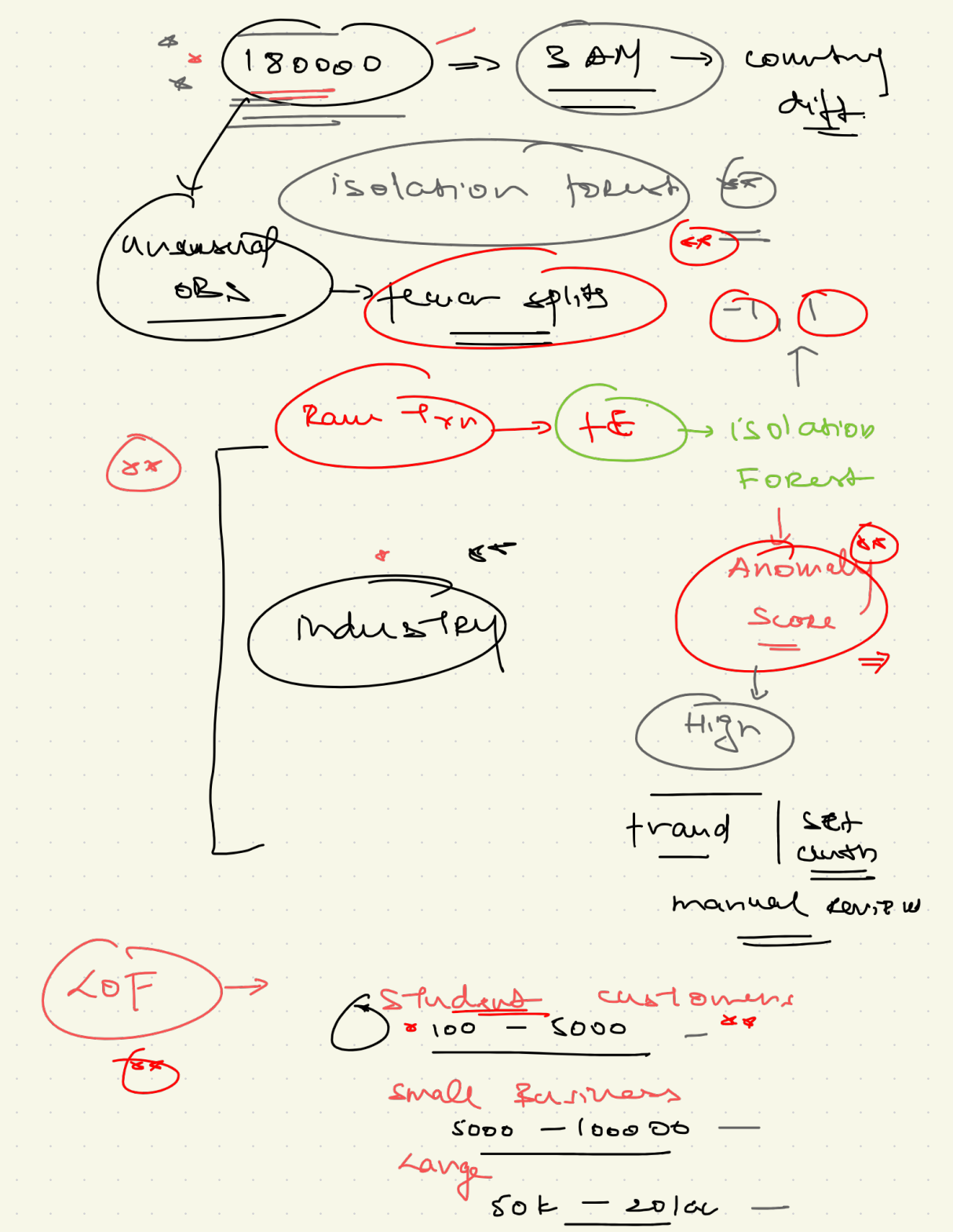
(**) e

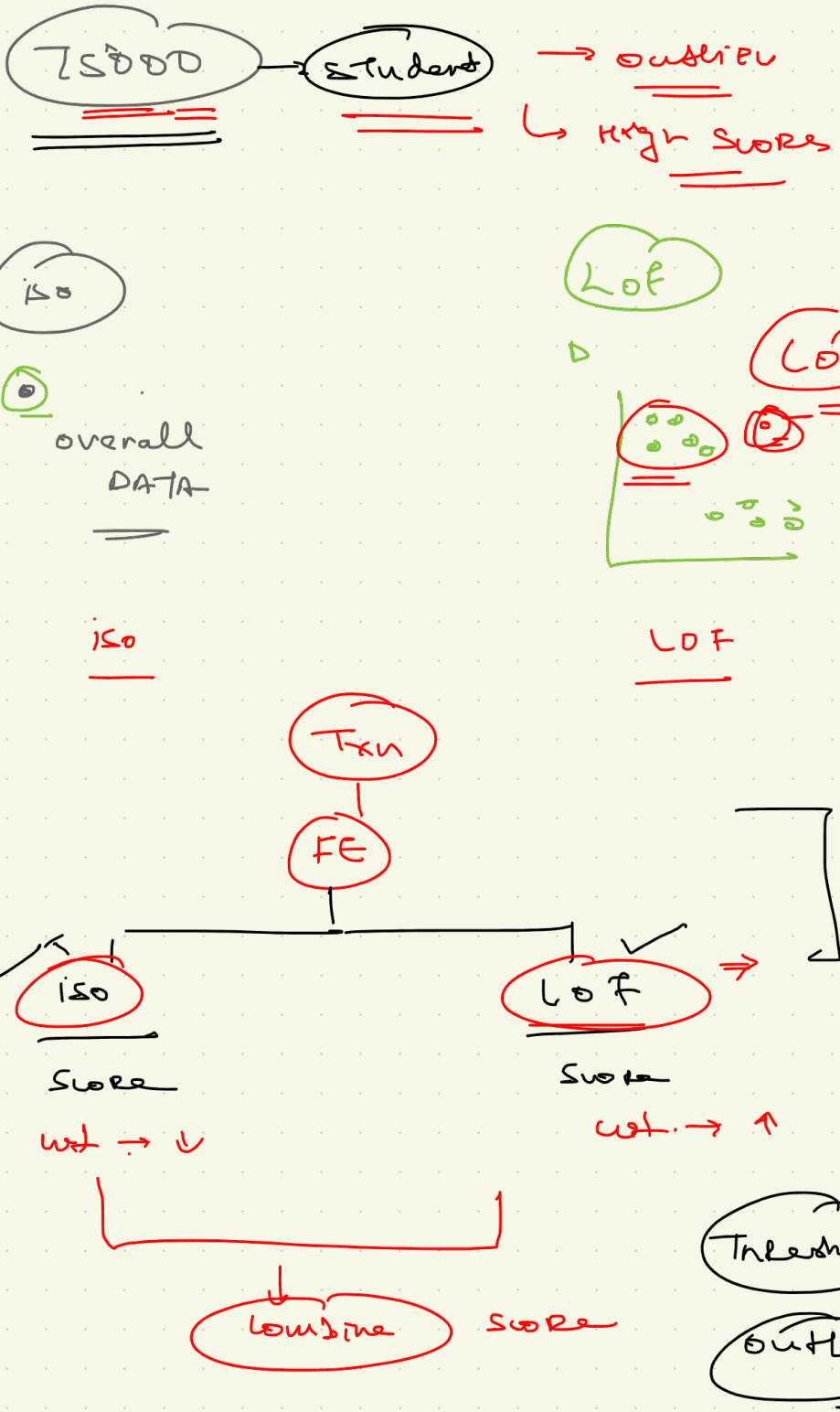
* 500 – 50000 RS

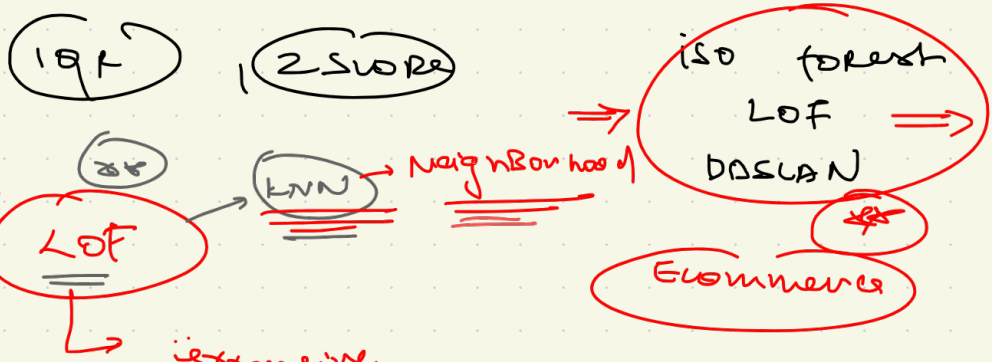
during the day

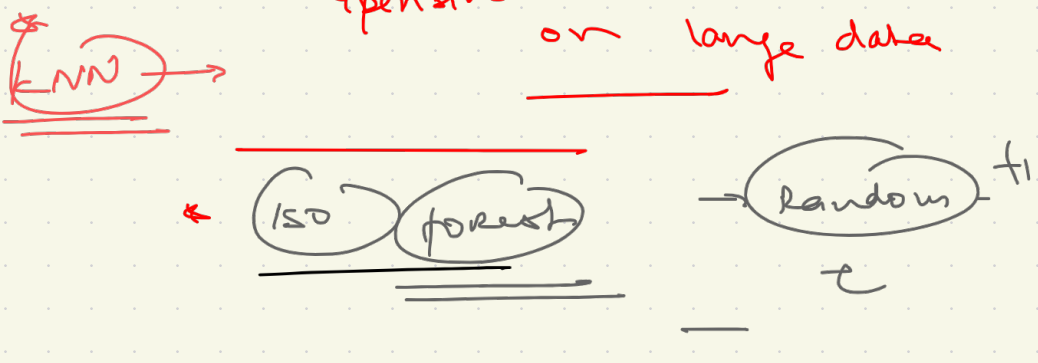
(*) amount, Time since prev. purchase, dist. from actual loc^n

if country mismatch


180000
SAM
country drift
isolation forest
unusual obs
fewer splits
-1
1
Raw Txn
FE
isolation Forest
Anomaly Score
High
industry
fraud
set aside
manual review
LOF
Student customers
100 — 5000
small farmers
5000 — 100000
Large
50k — 20lac


75000
Student
→ outlier
↳ High scores
iso
overall
DATA
LOF
LOF
iso
LOF
Txn
FE
iso
LOF
Industry
Score
wt → ↓
Score
wt → ↑
combine
score
Threshold
outlier




Problem Statement →

RMSE

MAPE

$$\frac{1}{n}\sum_{i=1}^{n}\left|\frac{\text{Actual} - \text{Predn}}{\text{Actual}}\right| \times 100$$

| Day | Actual | Predn | |
|---|---|---|---|
| 1 | 100 | 90 | 10% |
| 2 | 200 | 220 | 10 |
| 3 | 150 | 135 | 10 |

Day 1 → $\frac{(100 - 90)}{100} \times 100$

10%

$\frac{10 + 10 + 10}{3}$ = 10%

RMSE → 10000 ↓

MAPE ↓

goal → unsupervised → Tree

data → Handle missing values → Encoding → Scaling → X, Y

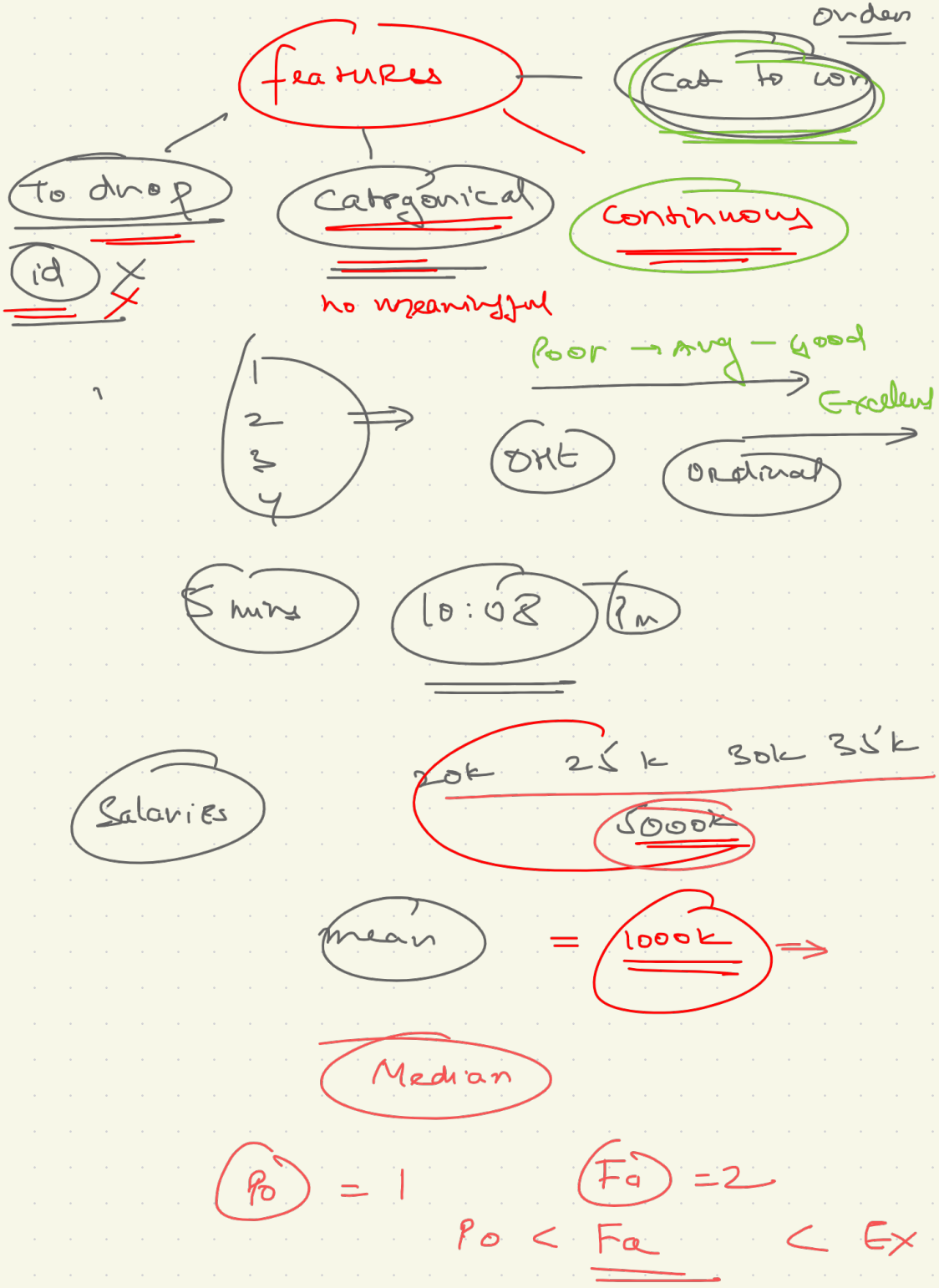
Shape

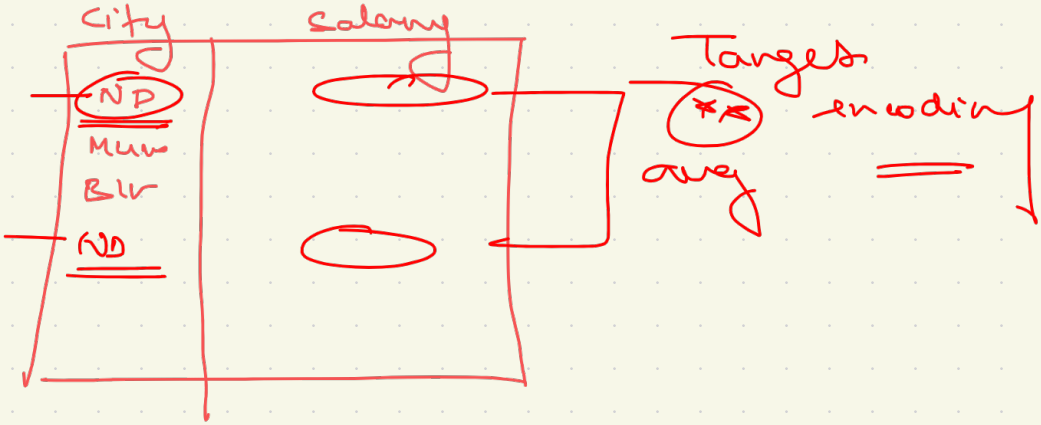
Type

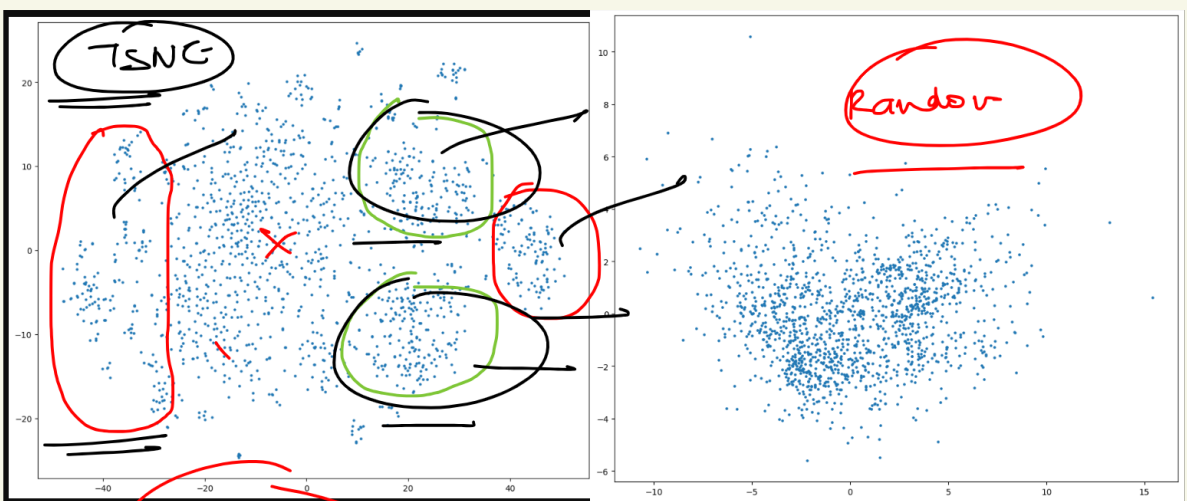
Baseline model → improve

100 – 200 → PCA

outliers


Order
features
Cat to con
to drop
Categorical
Continuous
id
no meaningful
1
2
3
4
Poor → Avg → Good
Excellent
OHE
Ordinal
5 mins
10:08
PM
Salaries
20k 25k 30k 35k
50000k
mean = 1000k
Median
Po = 1
Fa = 2
Po < Fa < Ex




TSNE

FE improve the scoring

Next Class

PCA

Dim Reduction TSNE

Results Clustering


Monday
Explore